

Smart Vision Assist



Supervised by
Prof. Dr. Walid Rabie
Dr . Yasmine Nagy

Team members



Aya Abdelhakam

Software Engineering

Khadija Mohamed

Software Engineering

Habiba Yehia

Computer Science

Hana Khamis

Software Engineering

Agenda

- Introduction
- Project Idea
- Literature review
- Analysis & Design
- Technologies that will be used
- Prototype and interfaces of the application
- Test plan
- Future tasks and implementation plan





1.Introduction

Motivation - Problem Statement

Introduction

Visual impairment affects people across all age groups and has a deep impact on daily life:

- It can affect **children, adults, and the elderly**, regardless of background.
- The condition ranges from **low vision to total blindness**.
- It directly impacts **safety, mobility, and independence**.
- Even simple daily tasks require **extra tools or support**, such as navigation assistance or guidance.





statistics

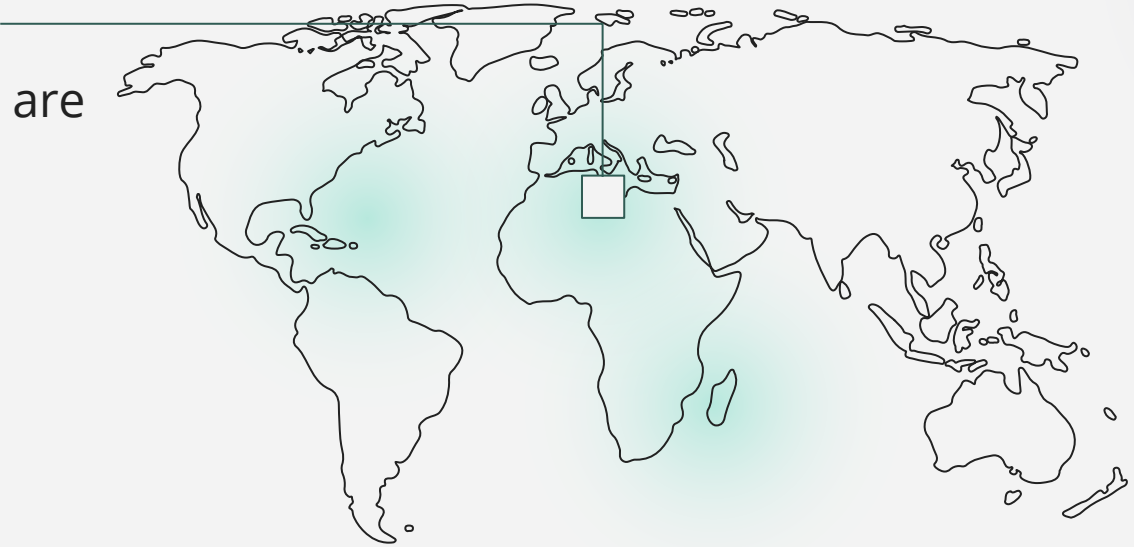
According to **WHO** There are

2.2 Million

Visually Impaired People in
Egypt

2.2 Billion

Visually Impaired People
Worldwide



Problem Statement

Problem 1: Safety Risks

Visually impaired individuals face a high risk of collisions, falls, and injuries due to limited environmental awareness.

Problem 2: Poor Visibility Conditions

Low-light environments make navigation significantly more difficult and increase the possibility of accidents.

Problem 3: Dependence on Others

Many users rely heavily on caregivers or strangers for safe movement in unfamiliar environments.

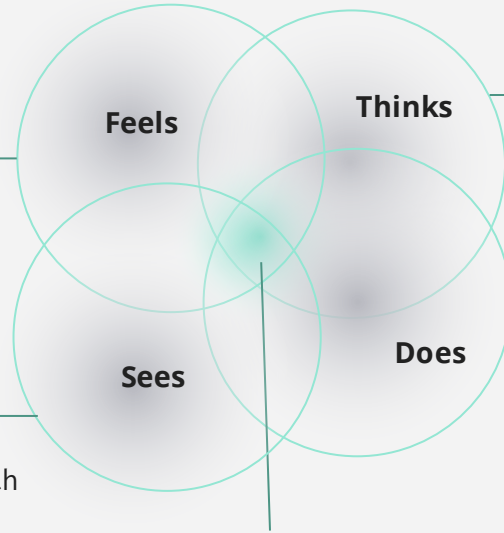


Empathy Map

Fear and anxiety about tripping or colliding with people or obstacles.

Embarrassment or discomfort when asking for help in public.

Crowded or busy public spaces with unpredictable movement.



“How can I move safely without bumping into obstacles?”

“Will people understand if I ask for assistance?”

- Frequently asks others for help in difficult situations.
- Avoids crowded or unfamiliar places

Visually Impaired Person



2. Project Idea

Project Aim - Objectives- Key Stakeholders-
Contributions

Project Aim

The aim of Smart Vision Assist is to develop an AI-powered mobile application that assists visually impaired individuals in navigating indoor environments safely and independently. The system uses Computer Vision and Deep Learning techniques to:

- Detect surrounding obstacles in real time.
- Estimate the relative distances of detected objects.
- Assess potential risks and dangerous situations.
- Provide immediate voice and vibration alerts to improve environmental awareness, mobility, and user safety.

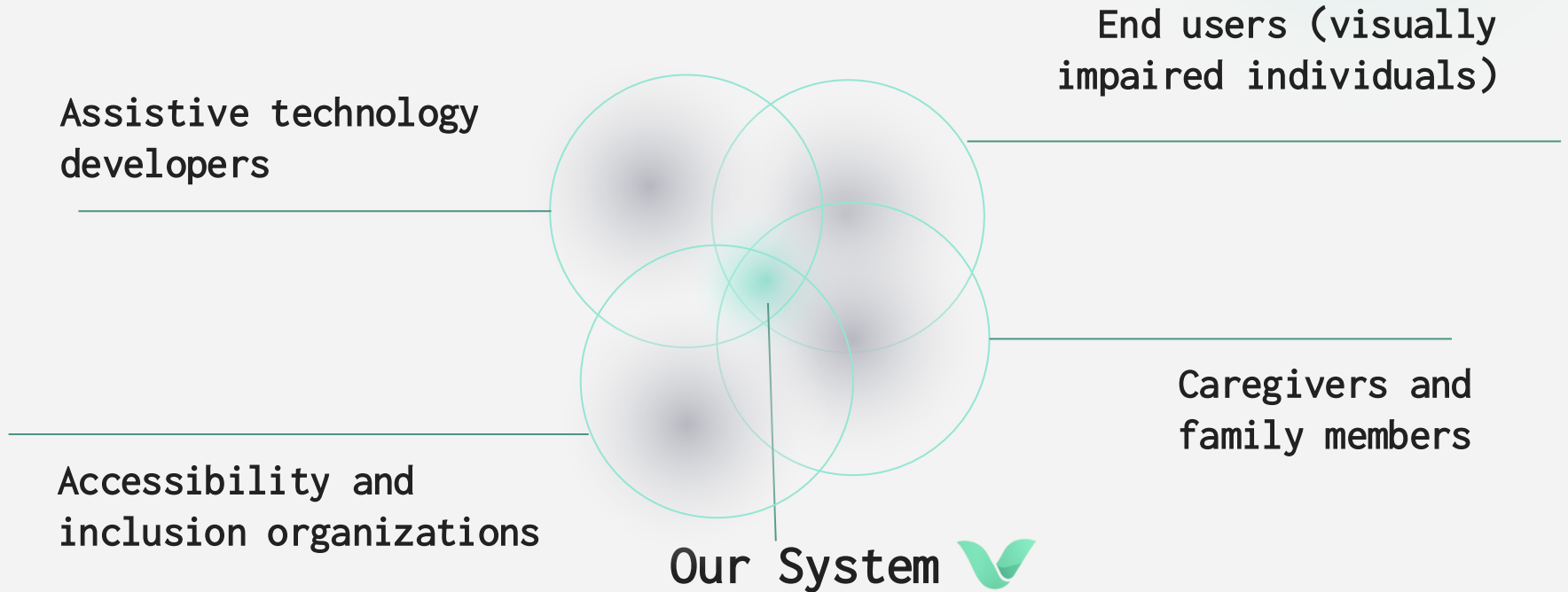


Objectives

- 1 Presents a smart mobile system for safer navigation
- 2 Combines data for improved environmental perception
- 3 Enables real-time risk assesment and safe path guidance
- 4 Delivers directional feedback using audio & haptics
- 5 Focuses on reliability and real-time performance



Key Stakeholders



Contributions

- Increases **independence and mobility** for visually impaired individuals.
- Enhances **personal safety** in daily navigation, especially in crowded and risky environments.
- Reduces **social discomfort and reliance on others** in public spaces.
- Improves overall **quality of life** through safer and more confident movement.
- **Low-Cost** Smart Assistance: Eliminates the need for expensive specialized hardware by relying only on a smartphone camera and AI models.



3. Literature Review





SWOT Analysis

STRENGTH

- Real-time AI detection
- Voice & vibration alerts
- Low-cost solution
- Easy to use

WEAKNESS

- Limited detection accuracy
- High battery consumption
- Depends on camera quality



THREAT

- Rapid technology changes
- Competing assistive solutions
- Environmental limitations

OPPORTUNITY

- Growing assistive technology market
- Expansion to outdoor navigation
- Integration with smart devices
- Wider adoption in healthcare



Similar Systems



Overview

Several solutions already exist to support visually impaired individuals in understanding their surroundings and navigating safely. These systems vary in complexity, cost, and level of technological dependence.

- Low-Tech Solutions : Traditional tools that rely on physical or human assistance with minimal technology.
- High-Tech Solutions: AI-powered and sensor-based systems that provide digital, audio, or visual assistance.



Low-Tech Solutions

Examples:

- Guide dogs.
- Tactile paving (raised ground indicators)
- Braille signs and printed maps.
- Verbal assistance from people nearby.



Advantages

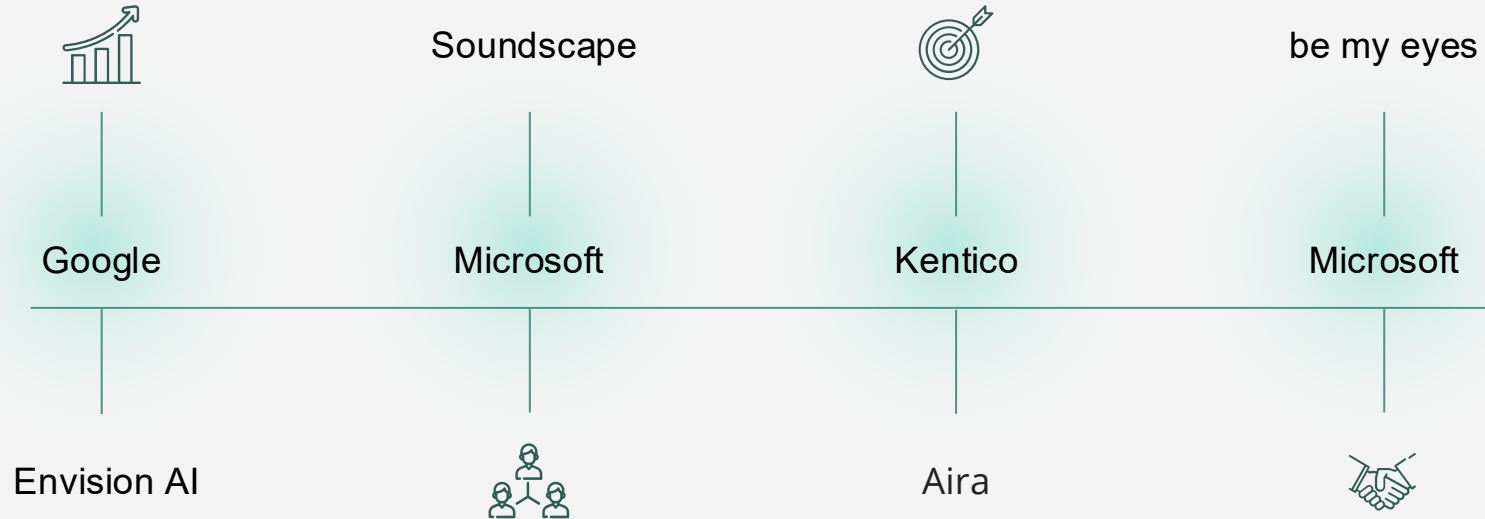
- Highly reliable and always accessible
- Low cost and widely available
- Easy to learn and use

Disadvantages

- Cannot detect dynamic or sudden obstacles
- Strong dependence on physical proximity
- Less effective in unfamiliar or complex environments



High-tech systems



Google Envision



Key Features

- Object Recognition
- Text Reading
- Scene Understanding

Limitations

- Performance may vary in different conditions
- Focused more on recognition than mobility assistance



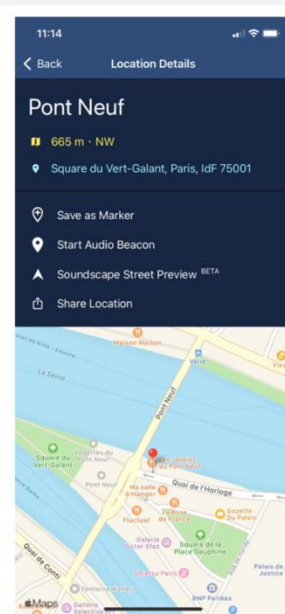
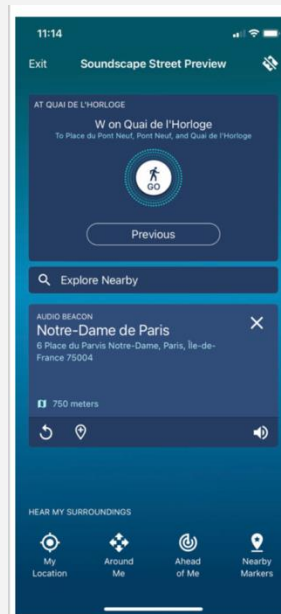


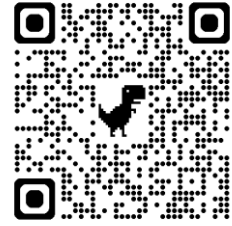
Key Features

- Spatial Audio Navigation
- Landmark Awareness

Limitations

- No Obstacle Detection
- No Depth Estimation



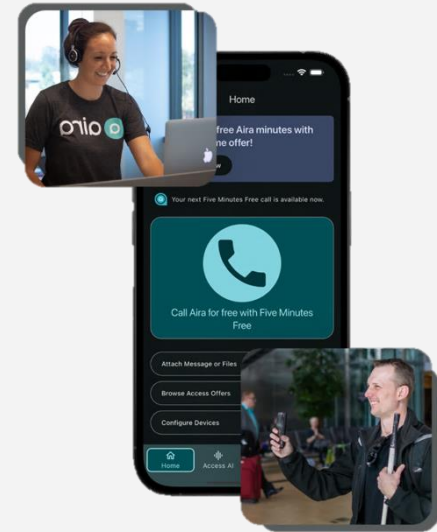


Key Features

- Live Human Assistance
- Effective in Complex Situations

Limitations

- Paid Subscription
- Internet Dependency



Be My Eyes

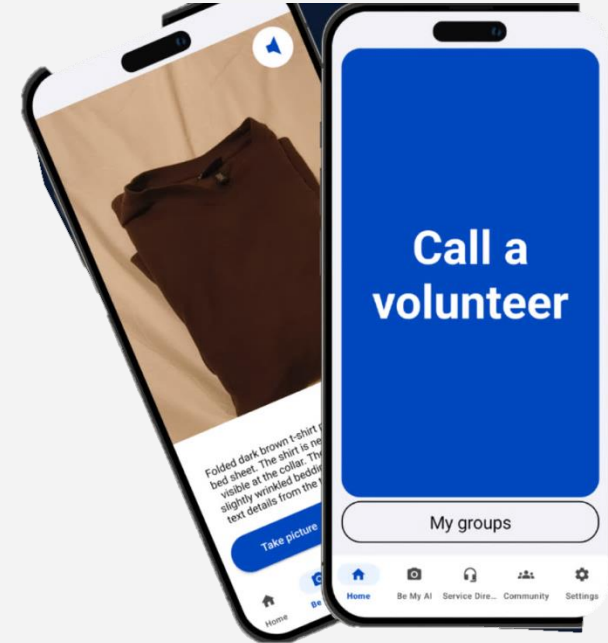


Key Features

- Volunteer-Based Assistance
- Easy to Use

Limitations

- No Autonomous Navigation
- Requires Available Volunteers



Ray-Ban Meta glasses



- Smart glasses designed for communication and media capture. They are not specifically intended for navigation assistance for visually impaired users.



@Lilymhe on instagram

Feature comparison



feature	Envision	Aira	soundscape	Our system
Dynamic Safe Corridor	Partial	Human-based	No	Yes
Collision Prediction	Limited	Human judgement	No	Yes
Autonomous Operation	Yes	No	Yes	Yes
Risk Scoring	No	Human-based	No	Yes
Multi-modal Alerts	Audio	Voice	Audio	Audio + Haptic

Market Gap

- Existing solutions focus on **either hardware or software**, but rarely provide a fully integrated system.
- assistive tools are often **single-function applications**.
- There is a lack of solutions that combine **real-time vision enhancement, safe navigation, and risk awareness** in one app.
- Visually impaired users need a system that works **independently, in real environments, and across different conditions**.



Business Opportunity

- Current assistive systems address **individual accessibility needs**, but no single solution combines them effectively.
- Users often rely on **multiple tools**, making navigation and awareness more complex.
- Our system delivers an **smart-application solution** that unifies navigation, object detection, and risk assessment .
- By centralizing all features into one ecosystem, the system **simplifies user experience** and increases independence.
- This represents a clear opportunity for a **comprehensive, accessibility smart app platform**.





4. System Analysis & Design



Functional & Non Functional Requirements

Functional Requirements

ID	Requirement
FR1	Real-time Image Acquisition
FR2	Obstacle Detection
FR3	Distance Estimation
FR4	Risk Assessment
FR5	Notification Generation
FR6	Audio & Haptic Feedback



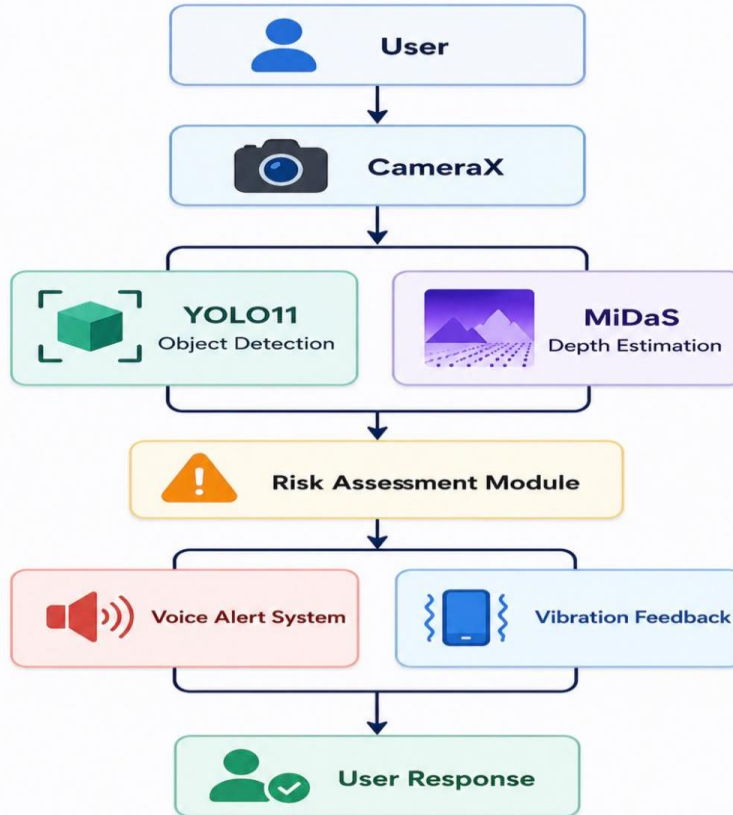
Functional & Non Functional Requirements

None-Functional Requirements

Category	Description
Performance	Low latency processing
Reliability	Continuous operation
Availability	Accessible whenever device is active
Usability	Accessible user interface
Portability	Android compatibility
Security	User data protection
Maintainability	Future scalability



System Architecture



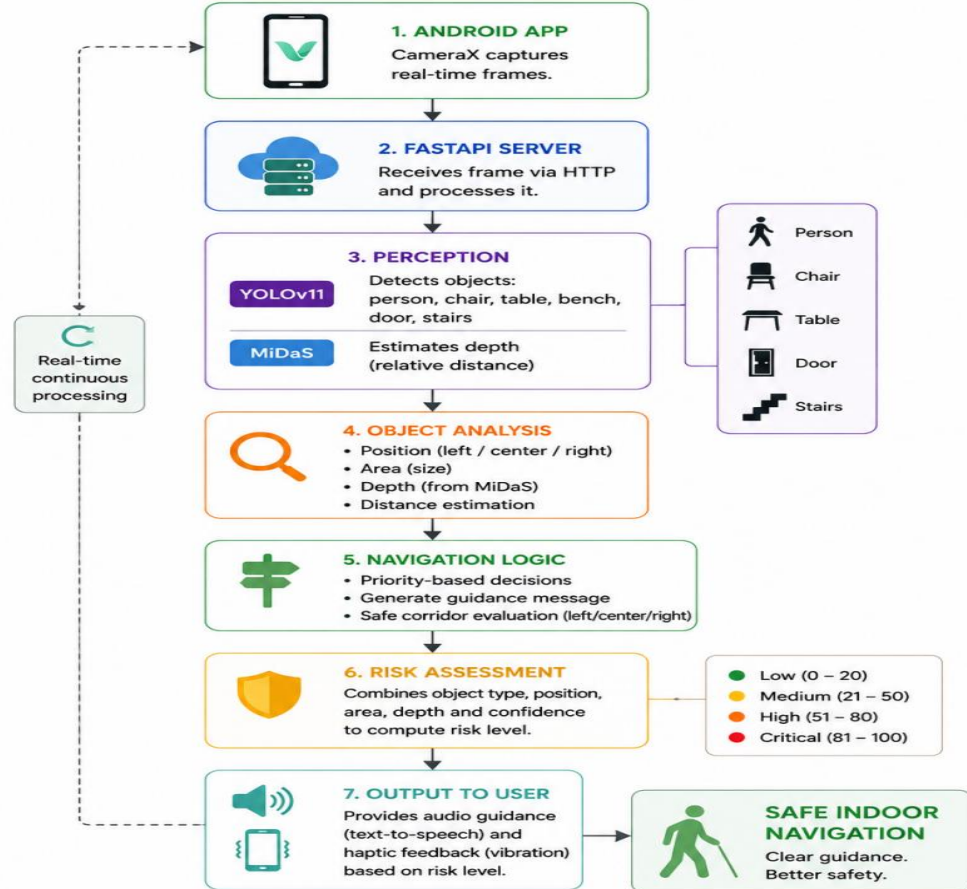
System workflow



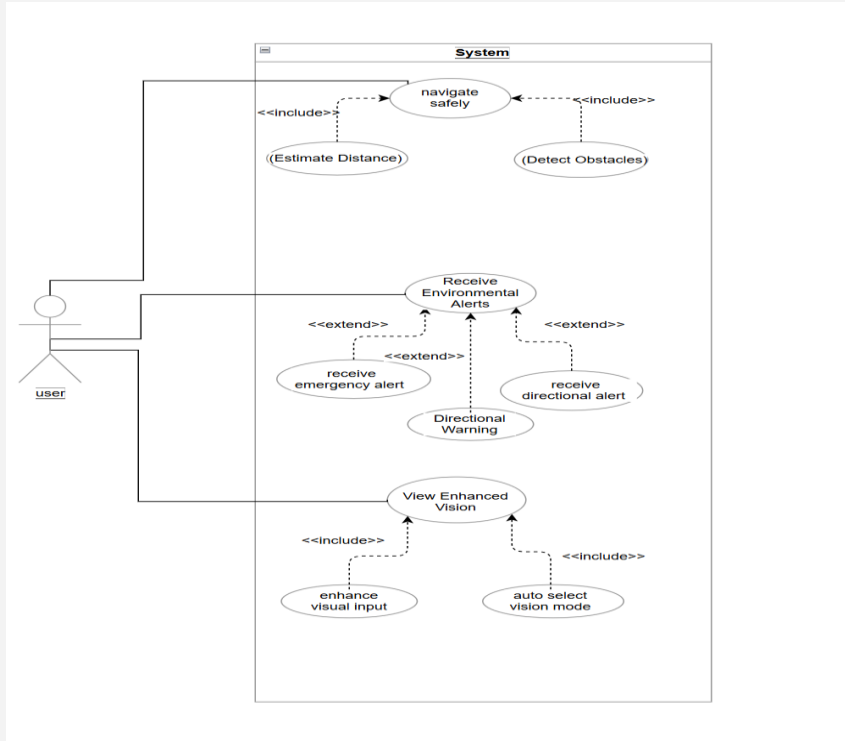
SMART VISION ASSIST

AI-Powered Indoor Navigation & Obstacle Avoidance

SYSTEM WORKFLOW

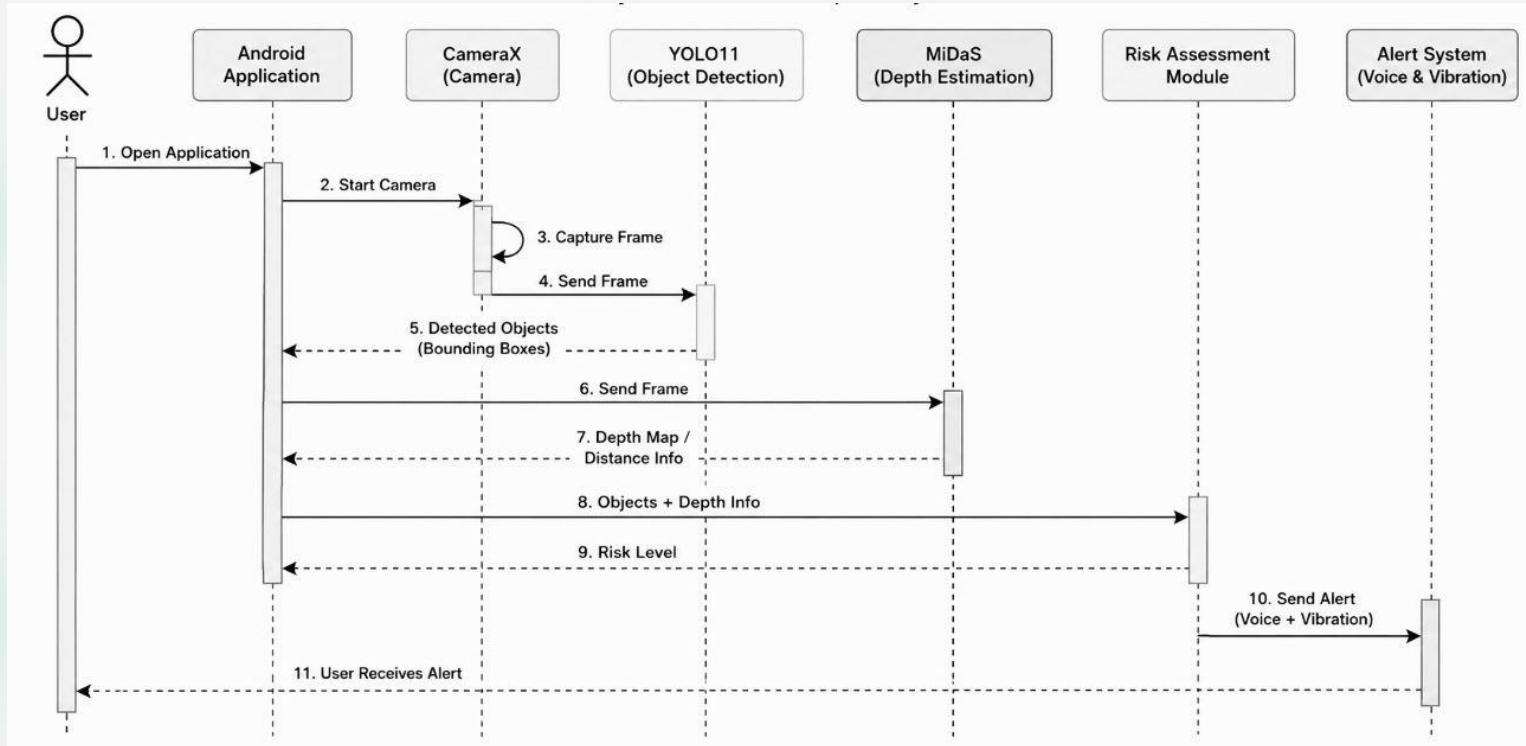


Use-Case Diagram



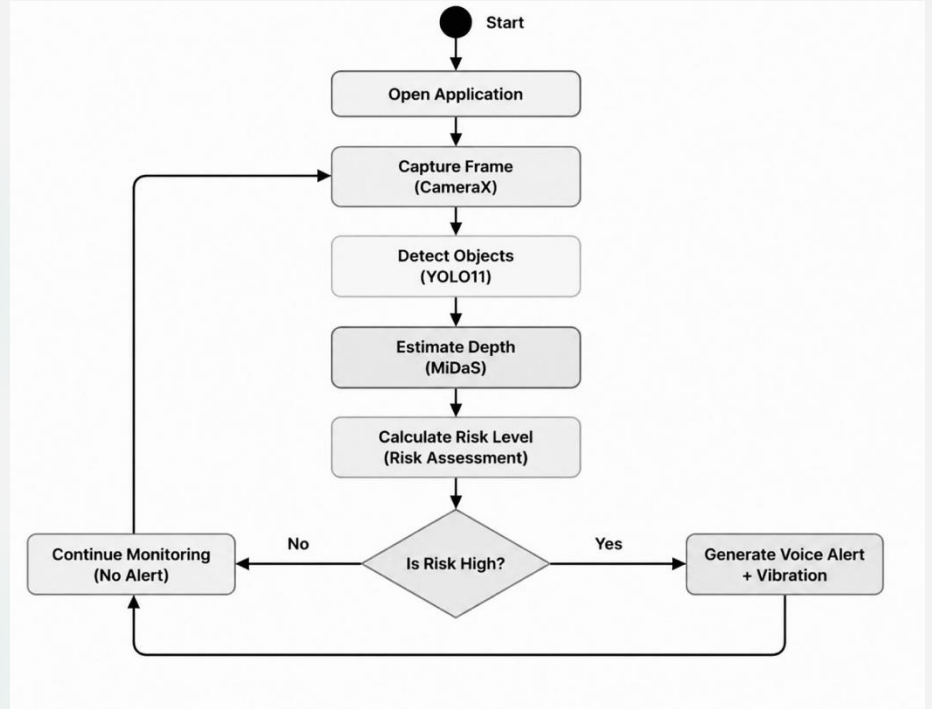
Sequence Diagram

Safety and motion perception



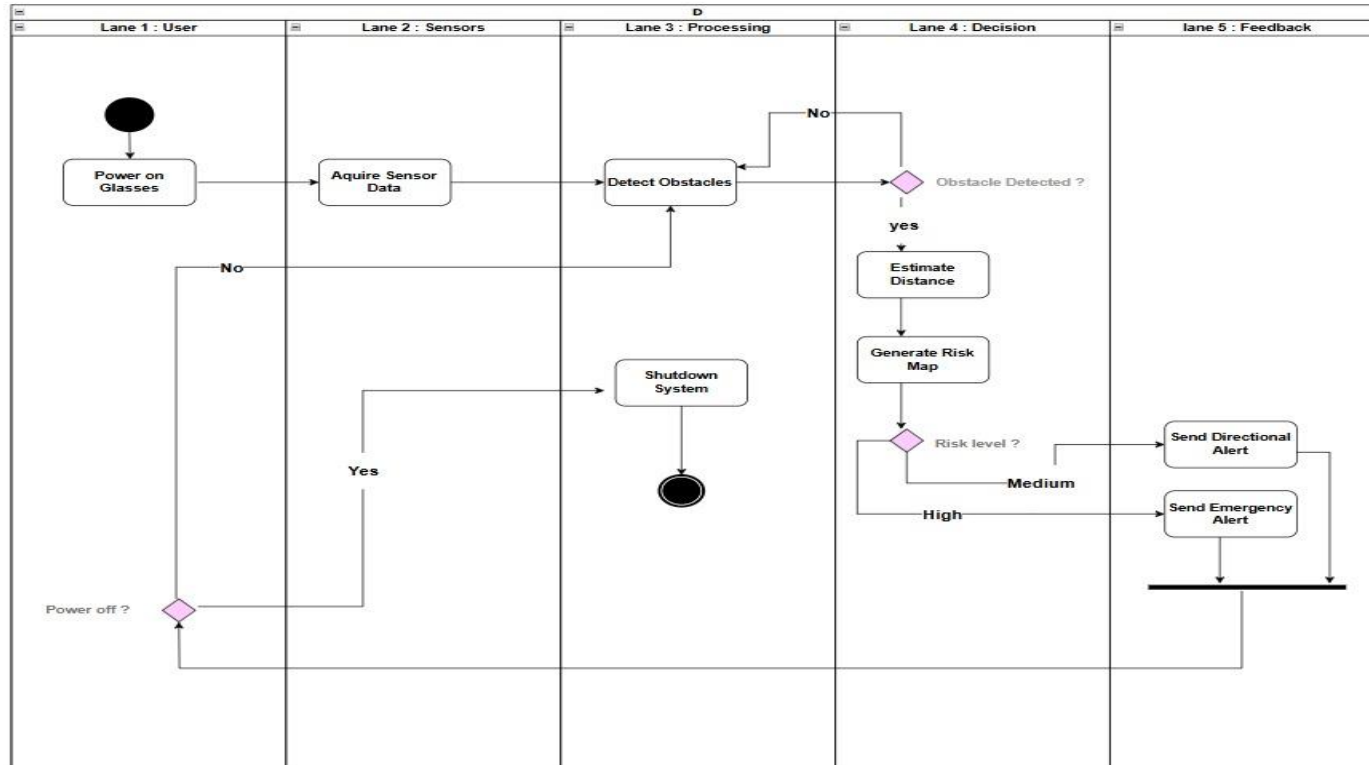
State Diagram

Safety and motion perception



Activity Diagram

Safety and motion perception





5. Used Technologies



FastAPI

**Authentication.
& Storage**



YOLOv11

**Real-time object
detection**



OpenCV

Image processing



PyTorch

Deep learning



JSON

Data exchange format



Figma

**Fast, collaborative
UI design and
prototyping**



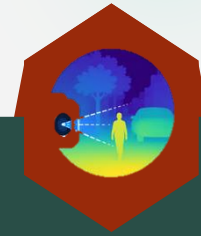
TTS

Voice notifications



Git

Version control



MiDaS

**Monocular depth
estimation**



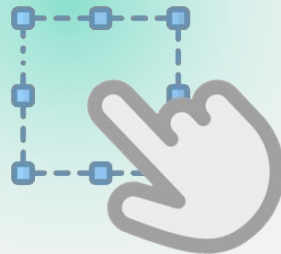
Android Studio

**Efficient and
streamlined app
development**

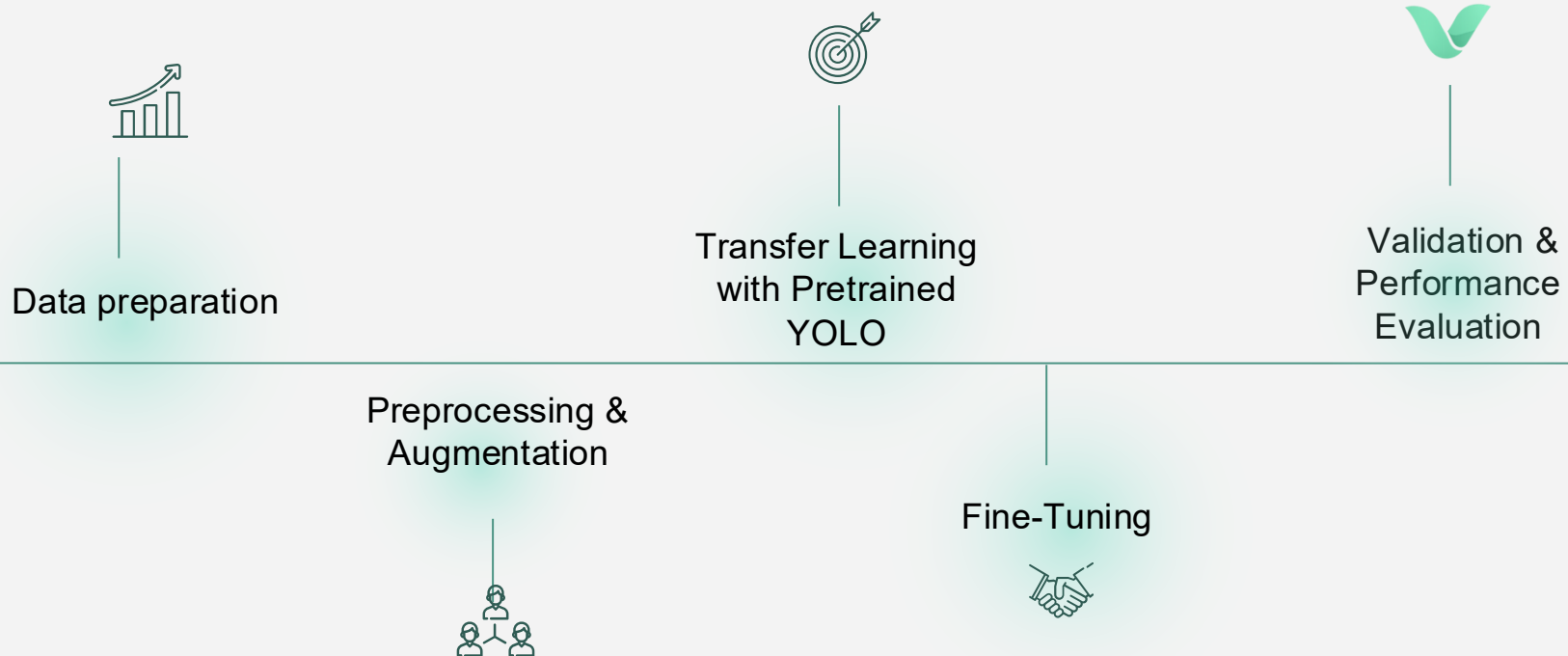
Dataset Selection

Why custom dataset?

- Existing datasets do not fully represent indoor navigation scenarios for visually impaired users .
- We needed navigation-specific classes such as :
 - Door
 - Stairs
- A combination of datasets was used to improve scene understanding .
- Our choice : COCO dataset + custom navigation dataset (doors , stairs , etc..)



Model Training Process





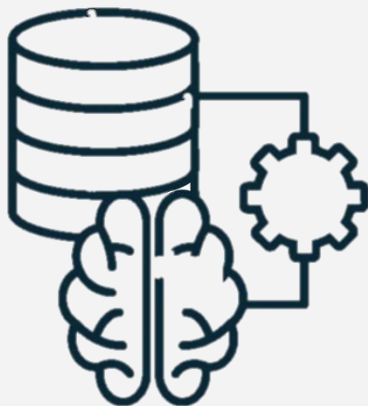
Preventing Overfitting and Underfitting

Overfitting Concerns

- Excellent training accuracy but poor real-world performance

Mitigation

- Data augmentation
- Validation monitoring
- Early stopping
- Diverse training samples



Underfitting Concerns

- Model unable to learn meaningful patterns .

Mitigation

- Additional training epochs
- Hyperparameter tuning
- Refinement of annotations
- Transfer learning from pretrained models

Risk Scoring Logic

Risk Score Computation

Each detected object is assigned a risk score based on the following factors:



Position

(Left / Center / Right)



Area

(Object size in image)



Depth

(Relative distance from MiDaS)



Object Type

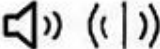
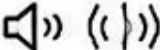
(Stairs, Person, Obstacle, Door)



Confidence

(Detection confidence)

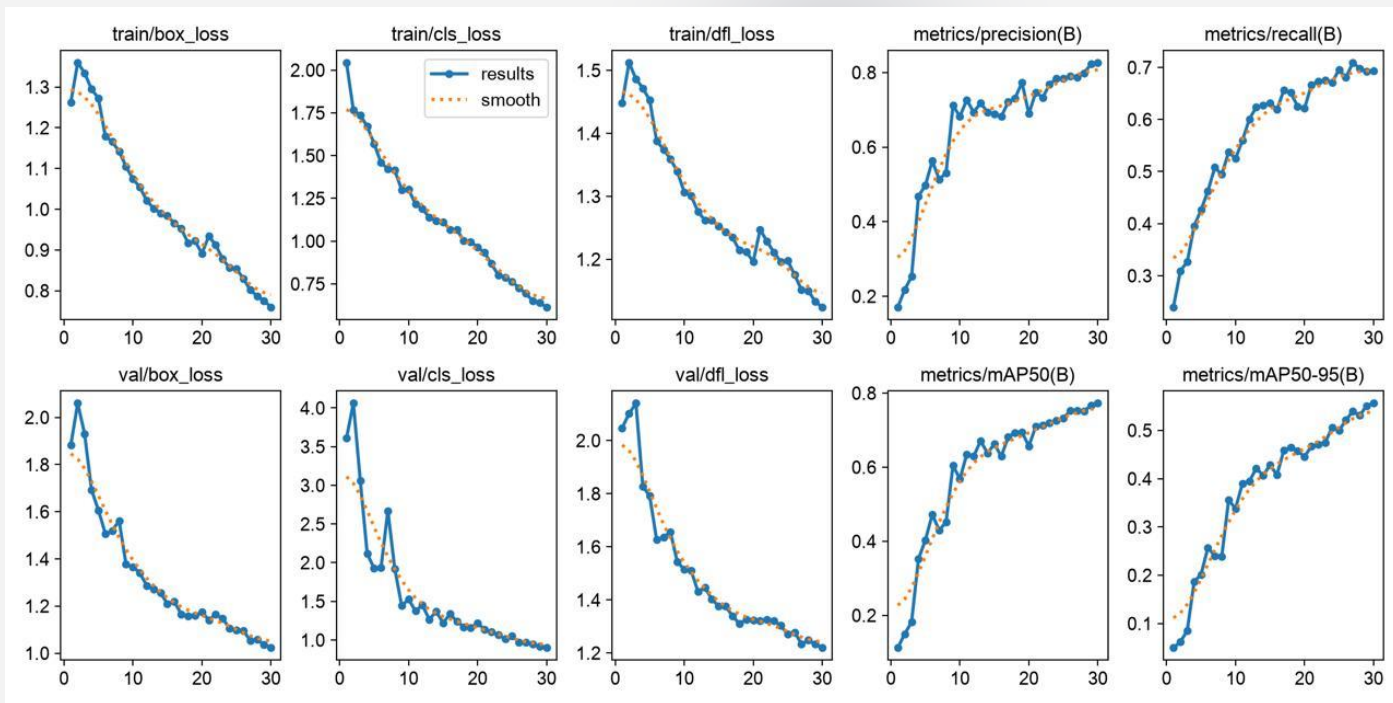
Risk Levels

 Low (0 – 20)	No feedback	
 Medium (21 – 50)	Speech warning only	
 High (51 – 80)	Speech guidance + short vibration	
 Critical (81 – 100)	Speech guidance + long vibration	

Source: Techatronic, 2023; Pratap et al., 2025

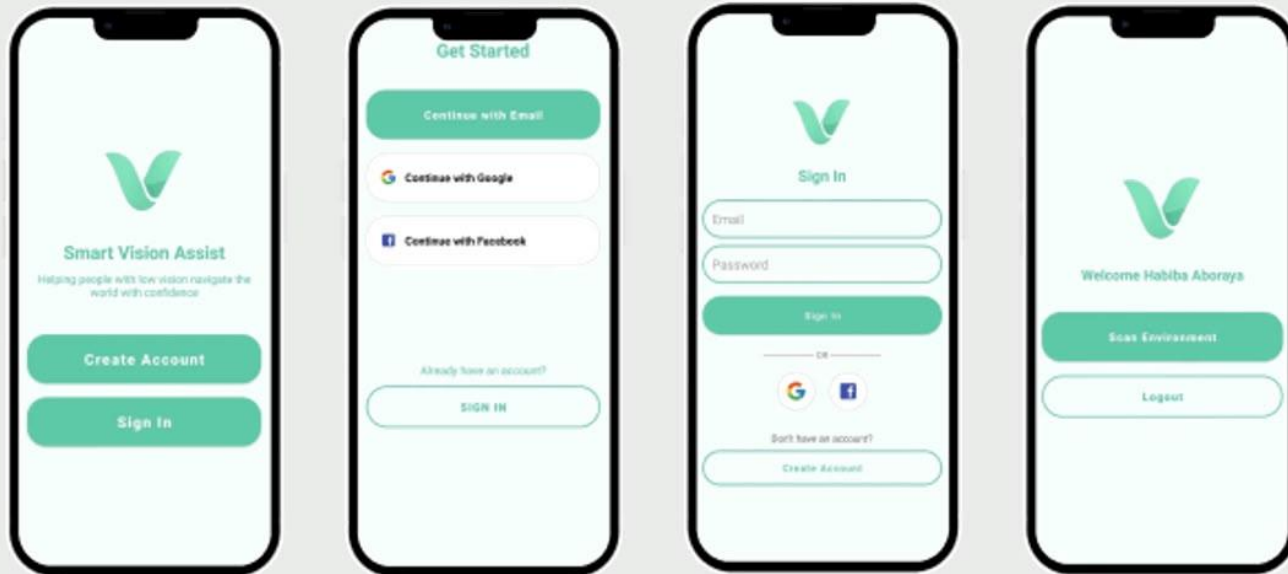


Detection Results



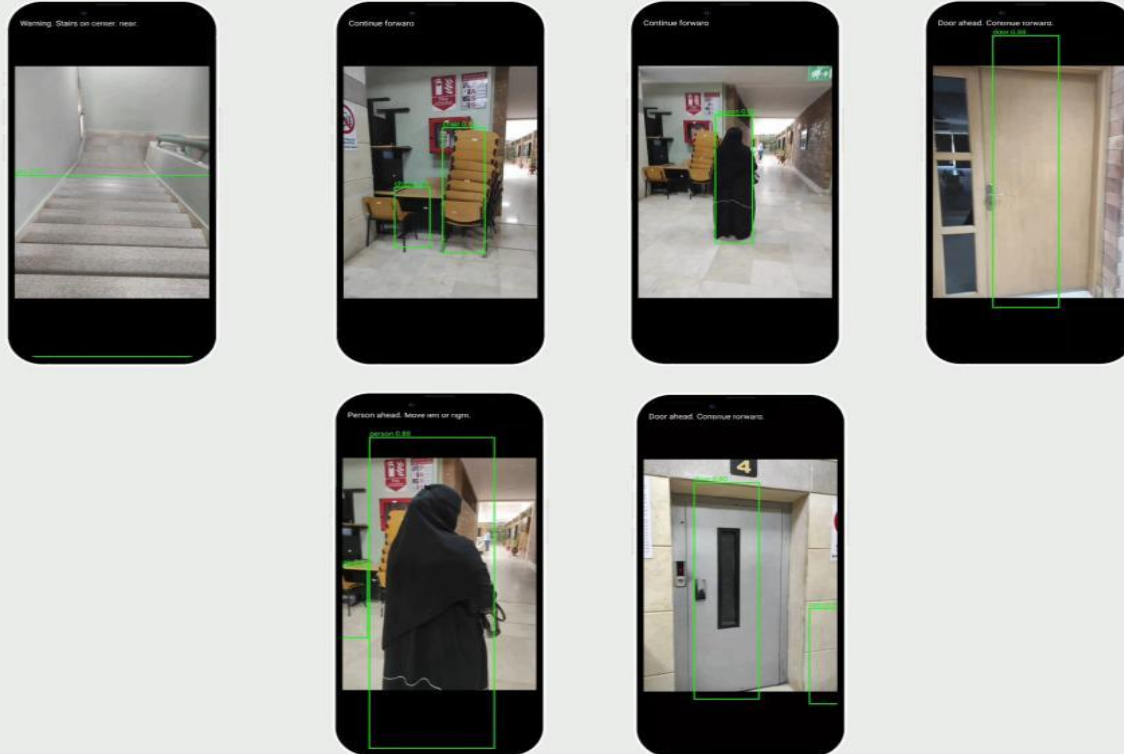
6.Mobile Application

LOGIN , SIGN UP INTERFACE

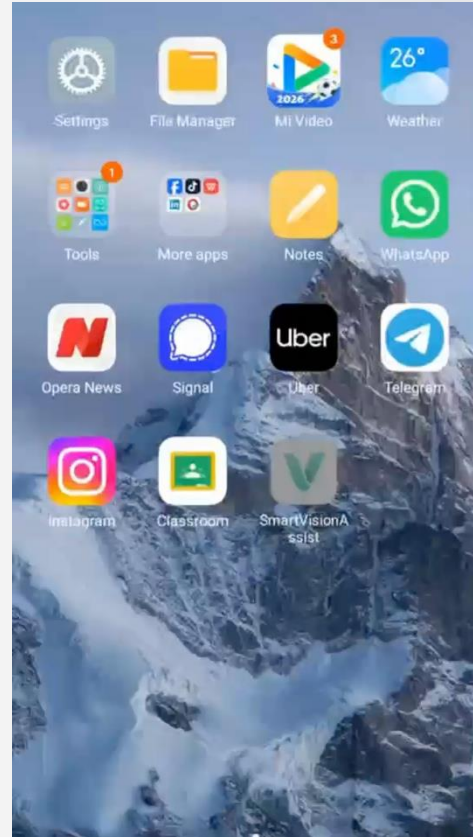


6.Mobile Application ✓

DETECTION AND LIVE ASSISTANCE INTERFACE



Prototype





7. Test plan & future tasks



Test Plan

- **Integration Testing** : Verified communication between the Android application, backend services, and AI models to ensure seamless data exchange.
- **System Testing** : Evaluated the complete workflow from camera frame capture to obstacle detection, risk assessment, and user notifications.
- **Performance Testing** : Measured response time and real-time processing capabilities under continuous camera streaming and multiple detection scenario.

Performed Test Samples



Test Case	Expected Result	Actual Result	Status
Object Detection	Detect surrounding objects correctly	Objects detected successfully	Passed
Depth Estimation	Estimate relative object distance	Relative depth estimated successfully	Passed
Risk Assessment	Identify potential hazards	Risk level generated correctly	Passed
User Alerts	Generate voice and vibration alerts	Alerts triggered successfully	Passed
System Performance	Process frames in real time	Stable real-time performance	Passed



8.Future plan



Near-Term
Implementation
Tasks



System
Optimization &
Expansion



Long-Term
Development
Roadmap



Future Plan

- Improve YOLOv11 detection accuracy.
- Enhance MiDaS depth estimation.
- Add OCR for reading signs and room numbers.
- Support outdoor navigation using GPS.
- Conduct real-world testing with visually impaired users.





9. Conclusion

- AI-powered mobile application for indoor navigation.
- Integrates YOLOv11, MiDaS, FastAPI, and Android.
- Detects obstacles and estimates relative depth in real time.
- Provides navigation guidance through voice and vibration feedback.
- Improves safety and independence for visually impaired users.





References

- [1] World Health Organization (WHO), "Blindness and Vision Impairment," World Health Organization, Geneva, Switzerland. [Online]. Available: <https://www.who.int/news-room/fact-sheets/detail/blindness-and-visual-impairment>. [Accessed: Jun. 2, 2026].
- [2] G. I. Okolo, T. Althobaiti, and N. Ramzan, "Assistive Systems for Visually Impaired Persons: Challenges and Opportunities for Navigation Assistance," *Sensors*, vol. 24, no. 11, Art. no. 3572, 2024.
- [3] D. Adapa, V. S. Shekhawat, A. Gautam, and S. Mohan, "Visual Assist System with Enhanced Localization for Indoor Navigation of Visually Impaired People," *IEEE Access*, 2024.
- [4] International Organization for Standardization (ISO), ISO 40500:2012—Information Technology—W3C Web Content Accessibility Guidelines (WCAG) 2.0, Geneva, Switzerland: ISO, 2012.
- [5] P. Pfreundschuh et al., "Sight Guide: A Wearable Assistive Perception and Navigation System for the Vision Assistance Race in the Cybathlon 2024," *arXiv preprint*, 2025.
- [6] IEEE Computer Society, IEEE Standard for Software and System Test Documentation (IEEE Std 829-2008), New York, NY, USA: IEEE, 2008.



Thanks

